

OFF ROAD BURNOUT - RULEBOOK

(SAE Collegiate Club, MMMUT Chapter)

1. Competition Overview

Off Road Burnout is an intercollegiate SAE event focused on RC car design, fabrication and off road racing.

The event encourages students to apply engineering concepts in a practical setting by fabricating and testing small-scale RC vehicles.

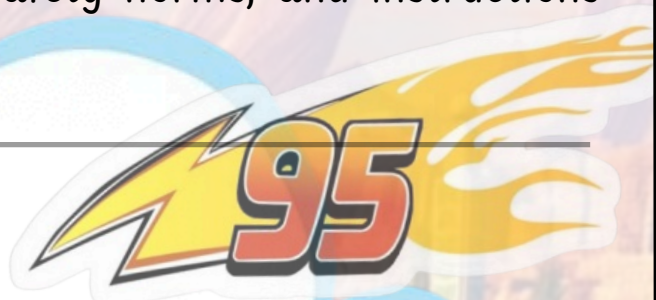
- Each team will fabricate its own RC car using the provided kit.
 - The event will run across three days including fabrication, testing, and racing.
 - The challenge is designed to promote teamwork, technical knowledge, and innovation under cost and time constraints.
 - Event Dates: 24th to 25th October 2026
 - Prize Pool: Rs. 15,000 + Medals + Exciting Goodies + Appreciation Certificates for Winners (All Finalists Also)
 - No registration fees for teams.
 - All teams will receive Participation certificates.
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2. Participation Requirements

- Each team must consist of a minimum of 2 and maximum of 5 members.
- Mixed teams (minimum 2 boys & 2 girls) are encouraged with extra 10 points.

- Every team, irrespective of team size, must pay ₹3000(minimal kit price).
- Teams must adhere to timelines, safety norms, and instructions given by coordinators.

1. EVENT DESCRIPTION



- **Orientation Day - Introduction & Material Distribution**

- o Theory and basic introduction about Mechanism.
- o Teams receive raw material and kit parts.
- o Presence of the Leader of each team is compulsory on the Orientation Day.

- **Day 1 - Chassis Fabrication, Technical Inspection & Dynamic Rounds**

- o Chassis fabrication begins.
- o Teams complete full RC car assembly.
- o Preliminary technical inspection conducted.
- o Manoeuvrability, durability, traction tests.

- **Day 2 - Pre-final race, Final race & Prize Distribution**

- o Head-to-head race among qualified teams.
- o Closing ceremony with results announcement.



4. Dynamic Events & Scoring

The competition consists of five dynamic stages. The maximum total score achievable is 315 points.

• 4.1 Technical Inspection (25 + 15 Bonus Points)

- RC car will be checked by event judges.
- Must comply with guidelines shared on Orientation Day.
- **25 points** for clearing inspection in first attempt.
- **20 points** for clearing inspection in second attempt.
- **+15 bonus** points for innovation/technology (within Rs. 1,000 budget).
- Each team must present valid GST bills of the innovation done in the car.
- Each team has maximum 2 attempts.

• 4.2 Manoeuvrability Test (50 Points)

- Teams must drive their RC car in an “8-shaped” track.
- **50 points** for successful completion in first attempt.
- **40 points** for successful completion in second attempt.
- Penalties:
 - o **-5 points** if car leaves the track.
 - o **-5 points** if car stops moving on track.
 - o **-1 point** per minute (Max 10 min time allowed) if any car repair is required (valid only for second attempt)
 - o After 10 minutes of car repairing, your team is disqualified
- **2 attempts maximum allowed.**



• 4.3 Durability Test (50 or 25 Points)

- Cars must jump across a ramp.
- All 4 wheels must land successfully after the jump, and the car should be in working conditions without any physical damage.
- If there is any circuit failure after the jump, the car can be qualified after circuit maintenance.
- If team fails in their circuit maintenance, team will be disqualified.
- **50 points** – for first attempt.
- **35 points** – for second attempt.
- **2 attempts maximum allowed.**

• 4.4 Pre-final Race (100 to 35 Points)

- Teams race head-to-head on off-road track.
- 12 or 15 teams will be selected for the pre-final race, and the race will be in a group of 3 teams respectively.
- Teams will be qualified for the final race, on point-distribution basis.
 - o **-5 points** if car leaves the track.
 - o **-5 points** if car stops moving on track.
- Points distribution:
 - o 1st Place – 100 pts
 - o 2nd Place – 75 pts
 - o 3rd Place – 55 pts
 - o 4th Place – 35 pts



• **4.5 Final Race (75 to 25 Points)**

- Further Qualified teams race head-to-head on off-road track.
 - 4 or 5 teams which qualify the pre-final race, will be selected for the final race.
 - No penalty of points in the final race.
 - Points distribution: (if 4 teams are qualified for final race)
 - o 1st Place - 75 pts
 - o 2nd Place - 60 pts
 - o 3rd Place - 45 pts
 - o 4th Place - 30 pts
 - Points distribution: (if 5 teams are qualified for final race)
 - o 1st Place - 75 pts
 - o 2nd Place - 60 pts
 - o 3rd Place - 45 pts
 - o 4th Place - 30 pts
 - o 5th Place - 20 pts
 - Finalists will be announced on points basis and all finalists will receive the Appreciation certificates.
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• **5. Winner Declaration**

- The team with the highest cumulative score from all rounds will be declared as the winner.
- In case of tie, race result will be considered as the winner deciding factor.
- Maximum achievable score = 325 points.



**See you at the starting
line, rookie.**